

Tandberg TDV-2200 Character Processing Functions and Variables

Summary of Renamed Functions and Memory Variables

I've analyzed and renamed all functions and memory variables related to incoming character processing in the Tandberg TDV-2200 terminal firmware.

Character Processing Functions Renamed

Main Character Input Processing

Address	Original Name	New Name	Function
0x1BE3	FUN_ram_1be3	process_incoming_character	Main character processing dispatcher
0x19CE	FUN_ram_19ce	call_escape_sequence_handler	Execute escape sequence via function pointer
0x1039	FUN_ram_1039	process_character_for_display	Process character for display output
0x1C97	FUN_ram_1c97	output_character_to_screen	Write character to screen buffer

Escape Sequence and Parameter Processing

Address	Original Name	New Name	Function
0x1F69	FUN_ram_1f69	parse_hex_parameter_sequence	Parse hexadecimal parameters
0x1F85	FUN_ram_1f85	validate_hex_digit_input	Validate hex digit characters
0x1F9A	FUN_ram_1f9a	read_and_normalize_input_character	Read and normalize input
0x11CE	FUN_ram_11ce	handle_special_control_sequence	Handle special control codes

Display and Output Functions

Address	Original Name	New Name	Function
0x1D95	FUN_ram_1d95	terminal_character_display_handler	Main terminal display handler
0x1C66	FUN_ram_1c66	calculate_cursor_position	Calculate cursor position
0x1E58	FUN_ram_1e58	output_character_with_spacing	Output character with spacing
0x1081	FUN_ram_1081	select_character_generator_bank	Select character set bank

Input Buffer and String Processing

Address	Original Name	New Name	Function
0x1E14	FUN_ram_1e14	process_input_string_buffer	Process input string buffer
0x1E48	FUN_ram_1e48	scan_and_process_string_characters	Scan and process string
0x1D49	FUN_ram_1d49	validate_cursor_row_range	Validate cursor row
0x1D62	FUN_ram_1d62	return_parameter_value	Return parameter value

Character Attributes and Control

Address	Original Name	New Name	Function
0x1EEF	FUN_ram_1eef	apply_character_attributes	Apply character attributes
0x20BF	FUN_ram_20bf	set_character_attribute_bits	Set attribute bits
0x11BF	FUN_ram_11bf	decrement_character_counter	Decrement character counter

Terminal Control and Communication

Address	Original Name	New Name	Function
0x0D1E	FUN_ram_0d1e	check_terminal_transmit_ready	Check terminal TX ready

Memory Variables Renamed

Character Processing Control Variables

Address	Original Name	New Name	Purpose
0x5E08	DAT_ram_5e08	escape_sequence_handler_pointer	Function pointer for escape processing
0x5E20	DAT_ram_5e20	character_generator_select_flag	Character generator bank selection
0x5E26	DAT_ram_5e26	display_output_control_flag	Display output enable/disable
0x5E27	DAT_ram_5e27	saved_display_mode_bits	Saved display mode state
0x5F16	DAT_ram_5f16	terminal_output_enable_flag	Terminal output enable flag

Display and Character Attributes

Address	Original Name	New Name	Purpose
0x5EF9	DAT_ram_5ef9	character_display_mode	Character display mode control
0x5DF6	DAT_ram_5df6	character_attributes	Current character attributes
0x5DF9	DAT_ram_5df9	display_update_flags	Display update control flags
0x5E02	DAT_ram_5e02	character_attribute_base	Base character attributes

Input Buffer Management

Address	Original Name	New Name	Purpose
0x5DFC	DAT_ram_5dfc	input_buffer_pointer	Input string buffer pointer

Character Processing Architecture

Main Character Input Flow

```
c
// Primary character processing entry point
Load_LH_FromRAM_5C2A() {
    (*serial_function_pointer)(); // 0x5C2A function pointer
}

// Main character processing dispatcher
process_incoming_character(uint8_t character) {
    if (character == 0xFC) {
        call_escape_sequence_handler(); // Execute via 0x5E08 pointer
    }
    else if (character == 0xCF) {
        // Read sequence parameters
        do {
            character = try_read_serial_data();
        } while (condition);
    }
    else if (character >= 0xB1 && character <= 0xB4) {
        // Handle special control sequences
        handle_special_control_sequence();
    }
    else if (character >= 0x20 && character <= 0x7F) {
        // Normal printable characters
        process_character_for_display(character);
        output_character_to_screen(character);
    }
}
```

Escape Sequence Processing

```
c
// Escape sequence handler dispatcher
call_escape_sequence_handler() {
    if (escape_sequence_handler_pointer != NULL) {
        (*escape_sequence_handler_pointer)();
    }
}

// Parameter parsing for escape sequences
parse_hex_parameter_sequence() {
    digit1 = validate_hex_digit_input();
    digit2 = validate_hex_digit_input();
    return (digit1 << 4) | digit2;
}
```

Display Output Processing

```
c
// Character display pipeline
output_character_to_screen(uint8_t character) {
    if (character >= 0x20 && character < 0x80) {
        // Normal character range
        calculate_cursor_position();
        select_character_generator_bank();
        write_to_display_buffer();
    } else {
        // Special character - display space
        output_character_with_spacing(0x20);
    }
}
```

Control Character Codes

Special Character Ranges

Code Range	Purpose	Handler
0x00-0x1F	Control characters	Special processing
0x20-0x7F	Printable ASCII	Direct display
0xB1-0xB4	Terminal control codes	handle_special_control_sequence
0xCE-0xCF	Parameter sequence markers	Parameter parsing
0xFC	Escape sequence trigger	call_escape_sequence_handler

Memory-Mapped Control

Variable	Function	Usage
escape_sequence_handler_pointer	Dynamic escape handling	Points to current escape processor
character_display_mode	Display mode control	Controls character rendering
character_attributes	Text attributes	Bold, blink, inverse, etc.
terminal_output_enable_flag	Output gating	Enable/disable terminal output

Terminal Capabilities

Supported Features

- 1. **Escape Sequence Processing:** Full escape sequence handler with function pointer dispatch
- 2. **Character Attributes:** Bold, blink, inverse, and other text attributes
- 3. **Cursor Control:** Position calculation and movement
- 4. **Character Sets:** Multiple character generator banks
- 5. **Input Processing:** Hex parameter parsing and validation
- 6. **Buffer Management:** String buffer processing and character accumulation

Processing Modes

- 1. **Normal Mode:** Direct character display
- 2. **Escape Mode:** Escape sequence processing
- 3. **Parameter Mode:** Multi-character parameter parsing
- 4. **Control Mode:** Special control character handling

This architecture demonstrates a sophisticated terminal with full escape sequence support, character attributes, and flexible character processing capabilities typical of professional terminals from the late 1970s/early 1980s era.